

Project Design Phase-II

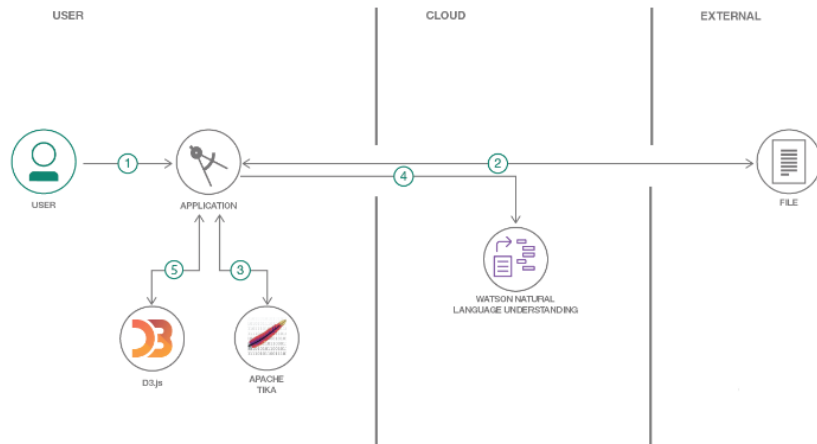
Data Flow Diagram & User Stories

Date	1 October 2026
Team ID	SWTID-2026-8255
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram shows how Incident information enters the ServiceNow system, is validated and controlled by UI Policies and Client Scripts, and is stored in the Incident record. The diagrams below are adapted specifically to this project.

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Supplied template reference

DFD Level 0 – Incident Management

Service Desk Agent	→	ServiceNow Incident Controls	→	Incident Record
Incident data Impact, Urgency, Assignment, State	→	UI Policy + Client Scripts • Mandatory / read-only control • Automatic Urgency update • Save validation • List-edit restriction	→	Validated Incident record
User changes	↔	Validation feedback / error messages	→	Saved / rejected submission

DFD Level 1 – Detailed Flow

Source / Process	Data Flow	Destination / Result
1. Service Desk Agent	Enters / changes Incident fields	2. Incident Form
2. Incident Form	Impact, Urgency, Assignment, State	3. UI Policy Evaluation
3. UI Policy	Impact = High condition	Assignment Group mandatory; Urgency read-only
4. onChange Client Script	Impact changes to High	Urgency automatically set to High
5. onSubmit Client Script	Impact = High + Assigned To empty	Submission rejected + validation message
6. onCellEdit Client Script	State edited from list	Change rejected + alert
7. Validated Form	All required rules satisfied	Incident record saved / updated

DFD Explanation

- The Service Desk Agent is the external entity that supplies Incident information.
- The Incident Form receives Impact, Urgency, Assignment and State values.
- The High Impact UI Policy controls field behavior when Impact is High.
- The onChange Client Script automatically sets Urgency to High for High Impact Incidents.
- The onSubmit Client Script prevents saving when Assigned To is empty for a High Impact Incident.
- The onCellEdit Client Script prevents direct State changes from the Incident list.
- When validation succeeds, the Incident record is saved; otherwise the user receives feedback and the change is rejected.

User Stories

The following user stories are written for the actual ServiceNow Incident project. Each story has a clear acceptance criterion, priority, and planned release.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Service Desk Agent	Incident Controls	USN-1	As a Service Desk Agent, I can create an Incident record with the required Incident information.	Incident can be submitted when required information is complete.	High	Sprint-1
Service Desk Agent	High Impact Control	USN-2	As a Service Desk Agent, I can have Assignment Group made mandatory when Impact is High.	When Impact = High, Assignment Group is mandatory; when the condition is false, the mandatory state is reversed.	High	Sprint-1
Service Desk Agent	High Impact Control	USN-3	As a Service Desk Agent, I can have Urgency made read-only when Impact is High.	When Impact = High, Urgency cannot be edited; when the condition is false, it becomes editable.	High	Sprint-1
Service Desk Agent	Automation	USN-4	As a Service Desk Agent, I can have Urgency automatically set to High when Impact changes to High.	Changing Impact to High sets Urgency to High and displays an information message.	High	Sprint-1
Service Desk Agent	Save Validation	USN-5	As a Service Desk Agent, I cannot save a High Impact Incident when Assigned To is empty.	Submit is blocked and a validation message identifies Assigned To as required.	High	Sprint-1
Service Desk Agent	List Control	USN-6	As a Service Desk Agent, I cannot change Incident State directly through list editing.	An alert is shown and the State value remains unchanged when edited from the list.	High	Sprint-2
Service Desk Agent	Form Update	USN-7	As a Service Desk Agent, I can change State from the Incident form.	State changes made through the form are saved successfully.	Medium	Sprint-2
Administrator	Testing & Validation	USN-8	As an Administrator, I can verify that UI Policies and Client Scripts work together as configured.	High Impact, missing Assigned To, successful save, reverse condition, list edit, and form update tests produce the expected results.	High	Sprint-2

Requirement Coverage Summary

Project Requirement	Covered By
High Impact UI Policy	USN-2, USN-3
Automatic Urgency	USN-4
Assigned To save validation	USN-5
State list-edit restriction	USN-6
Form-based State update	USN-7
End-to-end testing	USN-8

Note: The original template explains that a DFD should show how data enters, leaves, changes, and is stored, and its User Stories table uses user type, epic, story number, task, acceptance criteria, priority, and release. This completed document preserves that structure while replacing the sample mobile-registration content with the actual ServiceNow Incident project.